AeroPredict-AI: Predictive Maintenance System

AeroPredict-AI is a deep learning-based solution designed to estimate the Remaining Useful Life (RUL) of aircraft engines using multi-sensor temporal data.

🚀 Key Features

- **LSTM Architecture:** Captures temporal dependencies in engine degradation.
- **Normalization Pipeline:** MinMaxScaler pre-processing for high-variance sensor data.
- **Performance Evaluation:** Integrated RMSE and MAE calculation.

📊 Final Performance Metrics

- **Mean Absolute Error (MAE):** 16.53 cycles
- **Root Mean Squared Error (RMSE):** 27.22 cycles
- **Training Duration:** 100 Epochs

🛠 Usage

1. Run training: `python train_lstm.py`
2. Visualize and evaluate: `python visualize_results.py`